

Lab 2: Compressible Flow in de Laval Nozzles with Varying Pressures

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This experiment investigates compressible flow through a converging-diverging nozzle under different inlet and outlet pressure conditions. Pressure measurements were taken at several locations along the nozzle while varying the inlet pressure and outlet pressure independently. In these measurements, properties such as velocity, density, Mach number, and mass flow rate were calculated. The experiment was used to observe compressible flow behavior, including choked flow and shock wave formation inside the nozzle. The locations of shocks were found through experiments by looking at the pressure changes and then compared to what was expected based on isentropic flow rules and the relationship between area and Mach number. Results showed that as the pressure ratio increased, the flow approached choking at the throat, and shocks formed in the diverging section of the nozzle. Experimental trends generally agreed with compressible flow theory, although differences occurred due to experimental uncertainty, non-ideal flow effects, and measurement limitations.

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I. Nomenclature

A	= cross-sectional area of the nozzle
A^*	= critical throat area
a	= speed of sound
$\dot{m}$	= mass flow rate
M	= mach number
P	= static pressure
P_0	= stagnation pressure
P_1	= pressure at station 1 (nozzle inlet)
P_8	= pressure at station 8 (nozzle exit)
R	= specific gas constant for air
T	= temperature
T_0	= stagnation temperature
u	= fluid velocity
ρ	= density
γ	= ratio of specific heats
x	= position along the nozzle

II. Introduction

The purpose of this experiment is to investigate the behavior of compressible flow through a converging-diverging nozzle. The experiment examines the variations in pressure, temperature, velocity, and mass flow rate as air traverses the nozzle under varying inlet and outlet pressure conditions. By measuring these factors at different points along the nozzle, the experiment shows key behaviors of compressible flow, like how speed changes with the nozzle's shape, what happens during choked flow, and the formation of shock waves inside the nozzle. A converging-diverging nozzle as seen in **Fig. 1** is a device used to accelerate compressible fluid to high speeds, including supersonic velocities. The nozzle consists of three main sections, which are the converging section, the throat, and the diverging section. In the converging section, the cross-sectional area decreases and the velocity of the fluid increases while the flow remains subsonic. The throat is the smallest area in the nozzle, and it is the point where the flow can reach Mach 1 if it becomes choked. In the diverging section, the area increases. For compressible flow, once it reaches Mach 1 at the throat, the diverging section allows the flow to expand and accelerate further to supersonic speeds. This behavior is the opposite of incompressible flow and is what makes the converging-diverging nozzle essential for supersonic applications.

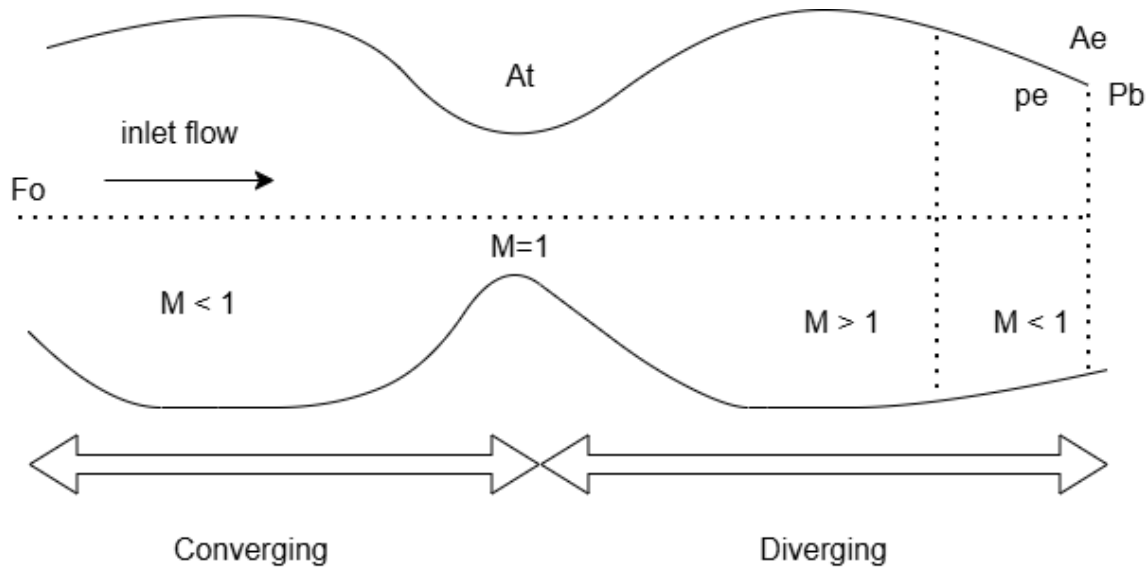


Figure 1. Converging-Diverging Nozzle.

In incompressible flow, the density of the fluid remains approximately constant even when pressure changes. This type of flow usually occurs at low velocities. In compressible flow, the density of the fluid changes significantly with pressure and temperature. We must consider additional effects like shock waves, temperature changes, and large pressure variations in compressible flow. This experiment studies a compressible flow, as the velocities within the nozzle approach or surpass the speed of sound.

The relationship between velocity and area in compressible flow depends on whether the flow is subsonic or supersonic. In subsonic flow ($M < 1$), decreasing the area of the flow passage causes the velocity to increase. This phenomenon is why the fluid accelerates in the converging section of the nozzle. However, for supersonic flow ($M > 1$), the opposite behavior occurs. Increasing the area of the passage causes the velocity to increase further. Therefore, once the flow becomes supersonic at the throat, the diverging section allows the gas to accelerate even more. This difference in behavior between subsonic and supersonic flow is a key concept in compressible aerodynamics.

When a shock occurs inside a nozzle, the flow transitions rapidly from supersonic to subsonic conditions. Across the shock, pressure and temperature increase sharply while velocity and Mach number decrease. Shock waves can appear in a converging-diverging nozzle when the outlet pressure is not low enough to maintain fully supersonic flow through the diverging section. In this case, a normal shock forms inside the nozzle and causes a sudden pressure rise. The location of this shock depends on the pressure ratio between the inlet and the outlet. While on the other hand, choked flow occurs when the flow velocity at the throat of the nozzle reaches the speed of sound, Mach 1. In this condition, the mass flow rate through the nozzle reaches its maximum possible value for the given inlet conditions. Once the flow is choked, decreasing the outlet pressure further does not increase the mass flow rate. This phenomenon occurs because disturbances downstream cannot travel upstream when the flow at the throat is sonic. Therefore, the throat area and the inlet conditions limit the flow rate. Rocket engines and high-speed propulsion systems commonly observe choked flow.

III. Procedures

The primary objective of this experiment was to identify and analyze the properties of a DeLaval nozzle. Figure 2 illustrates the specific converging-diverging nozzle that this experiment utilized.

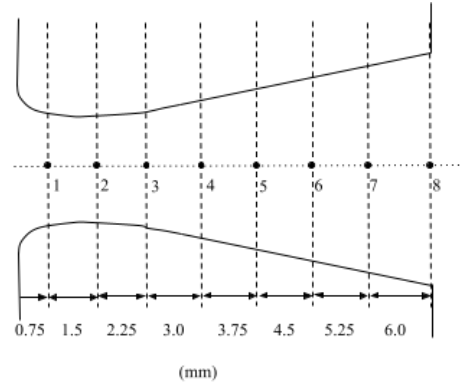


Figure 2. C-D Nozzle Diagram

Figure 2 illustrates the nozzle used in the experimental setup. The nozzle geometry consists of a converging section extending from a reservoir to the minimum cross-sectional area, referred to as the throat, followed by a diverging section that expands toward the exit area, which is the largest cross-section of the nozzle. Along the centerline of the nozzle, eight numbered pressure tap locations are installed. There is a separate pressure gauge for each of these places, and the instrumentation panel shows the readings for each one.

This nozzle assembly forms the central component of the experimental test rig. On the reservoir side (to the left of point one), the system is connected to a pressure supply line equipped with a regulator that allows the inlet pressure to be adjusted to a desired value. Similarly, the exit side of the nozzle (to the right of point eight) is connected to a pressure control system that allows the outlet pressure to be independently regulated.

At the nozzle exit, the flow passes through a rotameter, which is used to measure the mass flow rate of the air exiting the nozzle. The rotameter operates using a tapered tube containing a floating cone. The position of this cone varies according to the mass flow rate passing through the device. Higher mass flow rates cause the cone to rise within the tube, while lower flow rates cause it to descend. The cone height therefore provides a direct indication of the flow rate.

Additionally, two thermometers are included in the test rig to measure the air temperature at both the inlet and the outlet of the nozzle. Due to this instrumentation configuration, all experimental data were recorded manually in a spreadsheet during the experiment.

Before starting the experiment, it is important to measure and write down the conditions in the lab. These conditions include the atmospheric temperature and atmospheric pressure. These values are subsequently used to calculate the density of the air.

The air density can be determined using the ideal gas law. Because the Reynolds number is assumed to remain constant throughout the testing procedure, the ideal gas law can be rearranged to produce equation 2, where P represents pressure, T represents temperature, and R represents the specific gas constant. For air, $R = 0.2870$ kJ/kg·K.

$$PV = nRT \quad (2)$$

$$\rho = \frac{P}{RT} \quad (3)$$

The uncertainty associated with this density calculation can be determined using equation 4. In this equation, T represents temperature, P represents pressure, R represents the ideal gas constant, ΔP represents the pressure measurement uncertainty, and ΔT represents the temperature measurement uncertainty.

$$\delta\rho = \sqrt{\left(\frac{\delta P}{RT}\right)^2 + \left(\left(\frac{P\delta T}{RT^2}\right)\Delta T\right)^2} \quad (4)$$

The experimental procedure itself follows a systematic and repetitive process. The pressure control dial that is connected to the test rig's outlet sets the nozzle's outlet pressure at 50 kPa at first. After that, the pressure at the inlet is set to 100 kPa. Once steady conditions are reached, the mass flow rate indicated by the rotameter, the inlet temperature, the outlet temperature, and the pressures measured at the eight pressure taps are recorded in the data spreadsheet.

This procedure is repeated for inlet pressures of 200 kPa, 300 kPa, 400 kPa, and 450 kPa while maintaining the outlet pressure at 50 kPa.

In the second phase of the experiment, the inlet pressure is held constant at 450 kPa while the outlet pressure is varied. The outlet pressure is first set to 50 kPa, and the mass flow rate, temperatures, and pressure gauge readings are recorded. The outlet pressure is then increased sequentially to 100 kPa, 200 kPa, 300 kPa, and 400 kPa, with the corresponding measurements recorded for each condition. Once these measurements are completed, the data collection phase of the experiment is finished.

The primary objective of this experiment is to determine the parameters of compressible flow through a nozzle using the test rig. To accomplish this, several fundamental aerodynamic relations are applied. [2]

First, the conservation of mass is used to find the flow velocity at different nozzle points. The derivation begins with equation 5. The first term in equation 5 is eliminated because it represents a time-dependent component, and the flow is assumed to be steady and uniform. The integral in the second term becomes simpler by considering the velocity in the x-direction across the area of the nozzle. This results in equation 6, which can then be rearranged to determine the velocity at any point in the nozzle, as shown in equation 7. Because of the conservation of mass, the mass flow rate remains constant throughout the nozzle.

$$\frac{\delta}{\delta t} \int_{CV} \rho dV + \int_{CS} \rho(\mathbf{u} \cdot \mathbf{n}) d\mathbf{s} = 0 \quad (5)$$

$$\rho u_i A_i = \dot{m} \quad (6)$$

$$u_i = \frac{\dot{m}}{\rho A_i} \quad (7)$$

Next, the force acting on the nozzle is determined using the conservation of momentum. The flow is assumed to be one-dimensional, meaning that the velocity is considered only in the x-direction. This assumption simplifies the momentum equation considerably.

The general form of the momentum equation is given in equation 8. As before, the first term represents time-dependent effects and is eliminated because the flow is steady. The second term simplifies in a manner similar to the derivation used for the conservation of mass. The third term becomes the product of the flow area and the fluid density.

A key aspect of this analysis is accounting for the direction of the velocity vectors. The positive normal direction is defined as the direction from the inlet toward the outlet of the nozzle. Consequently, directional signs must be applied to the velocity, density, and area terms accordingly. Additionally, the only external force considered in this system is the one acting on the nozzle itself. The final simplified form of the equation is given in equation 9.

$$\frac{\delta}{\delta t} \int_{CV} \rho(\bar{u}) dV + \int_{CS} \rho \bar{u} (\bar{u} \cdot \hat{n}) ds + \int_{CS} \rho \cdot \hat{n} ds = \sum \overline{F_{ext}} \quad (8)$$

$$-\rho u_1^2 A_1 + \rho u_2^2 A_2 + \rho A_1 - \rho A_2 = F_{ext} \quad (9)$$

The next stage of data analysis involves applying Bernoulli's equation to determine the percent difference between the inlet and outlet conditions of the nozzle. In equation 10, the left-hand side will be referred to as LHS and the right-hand side will be referred to as RHS. The equation used here is a modified form of Bernoulli's equation, as there is no change in elevation throughout the nozzle.

$$P_1 + \frac{1}{2} \rho_1 u_1^2 = P_2 + \frac{1}{2} \rho_2 u_2^2 \quad (10)$$

$$\%Difference = \frac{|LHS-RHS|}{\frac{LHS+RHS}{2}} * 100 \quad (11)$$

Next, the Mach number and local speed of sound at various nozzle positions must be calculated. The Mach number represents the ratio of the fluid velocity to the speed of sound in the surrounding medium. The Mach number equation is shown in equation 12, while the speed of sound relationship is given in equation 13. [3]

$$M = \frac{u}{a} \quad (12)$$

$$a = \sqrt{\gamma RT} \quad (13)$$

Using the Mach numbers obtained from these calculations, the area-Mach number relation can be applied to determine the critical area of the nozzle, denoted as A^* . When this value is equal to the throat area, the flow is considered choked, meaning that the Mach number at the throat equals one. The area-Mach relation comes from mixing isentropic flow rules and shock equations, but it's often shown as a separate equation. The relation is given in equation 14, where the subscript i represents the properties at any location within the nozzle.

$$\frac{A_i}{A^*} = \left[\frac{1}{M_i} \left(\frac{2}{\gamma+1} \right) \left(1 + \frac{\gamma-1}{2} M_i^2 \right) \right]^{\frac{\gamma+1}{2(\gamma-1)}} \quad (14)$$

The isentropic flow relations, which relate the pressure and temperature at the reservoir to those at any point within the nozzle, are the next set of relations used in the analysis. For the purpose of this analysis, it is initially assumed that no shock waves occur within the nozzle and that the flow remains entirely isentropic. Under this assumption, these relationships can be applied throughout the entire nozzle. The pressure relation is given in equation 15, while the temperature relation is provided in equation 16.

$$\frac{P_0}{P_i} = \left[1 + \frac{\gamma-1}{2} M_i^2 \right]^{\frac{\gamma}{\gamma-1}} \quad (15)$$

$$\frac{T_0}{T_i} = 1 + \frac{\gamma-1}{2} M_i^2 \quad (16)$$

Although the analysis initially assumes that no shock occurs, the final parameter to be determined is the theoretical shock location within the nozzle. To determine this location, the flow from the reservoir to the point immediately before the shock is assumed to be isentropic. A normal shock then occurs, after which the flow from the shock location to the exit is again assumed to be isentropic. Using this approach, equation 16 provides the relationship used to determine the ratio between the stagnation pressure at the exit and the stagnation pressure at the reservoir. In this notation, $P_{1'}$ represents the pressure immediately before the shock, and $P_{8'}$ represents the pressure immediately after the shock.

$$\frac{P_{08}}{P_{01}} = \frac{P_{08}}{P_8} \frac{P_8}{P_1} \frac{P_1}{P_{01}} \quad (17)$$

$$\frac{P_{08}}{P_{01}} = \left[\frac{(\gamma+1)M_s^2}{(\gamma-1)M_s^2+2} \right] \left[\frac{\gamma+1}{2\gamma M_s^2 - (\gamma-1)} \right] \quad (18)$$

To find out where the shock is located in the nozzle, we take the Mach number (a dimensionless quantity representing the ratio of the speed of an object to the speed of sound) from equation 18 and plug it into the area-Mach relation in equation 14 to find the shock area. For a shock to occur inside the nozzle, the flow must first reach Mach 1 at the throat; therefore, the throat area is used for A^* .

By using the known diameter and the ratios of areas at different pressure tap locations, we can find out where the shock is located in the nozzle. Using the equations described above, six plots will be generated using a modeling program such as MATLAB. Three of these plots will be based on experimental data, and three will be based on theoretical predictions.

For the experimental plots, the first two will show the ratio of the pressure measured at each pressure tap to the atmospheric pressure plotted against the tap position in millimeters. One of these plots will correspond to the fixed inlet pressure case, while the second will correspond to the fixed outlet pressure case. The third experimental plot will display the mass flow rate as a function of the ratio of inlet pressure to outlet pressure.

Three theoretical plots will also be produced. The first will show the Mach number as a function of position along the nozzle. The second will show the ratio of the pressure at any pressure tap to the stagnation pressure plotted against the nozzle position. The third will show the ratio of the temperature at any point to the stagnation temperature plotted against the nozzle position.

In addition to these plots, four tables will be constructed. These tables will include the recorded experimental data, the calculated force acting on the nozzle, the results obtained from Bernoulli's equation, and the calculated shock formation parameters within the nozzle. Once these tables and plots have been completed, the data processing portion of the experiment will be finished.

IV. Results

A. Tables

Table 1. Experimental Data

	Variable Pressure (kPa)	Speed of Sound (m/s)		Density (kg/m ³)		Velocity (m/s)	
		Station 1	Station 8	Station 1	Station 8	Station 1	Station 8
Fixed Outlet Pressure (kPa)	146.73	138.73	154.73	343.787	343.787	1.6433	1.8328
	146.73	196.73	146.73	343.787	343.787	2.3303	1.7381
	146.73	261.73	131.73	343.787	343.787	3.1003	1.5604
	146.73	335.73	115.73	343.787	343.787	3.9769	1.3709
	146.73	356.73	86.73	343.787	343.787	4.2256	1.0274
Fixed Inlet Pressure (kPa)	546.73	356.73	86.73	343.787	343.787	4.2256	1.0274
	546.73	356.73	176.73	343.787	343.787	4.2256	2.0934
	546.73	366.73	276.73	343.787	343.787	4.3441	3.278
	546.73	356.73	384.73	344.3708	343.787	4.2113	4.5573
	546.73	436.73	478.73	344.3708	343.787	5.1557	5.6707

Table 2. Force on Nozzle

	Ratio of Inlet Pressure over Outlet Pressure	Force of Fluid on Nozzle (N)
Fixed Outlet Pressure (kPa)	1.3408	-0.3817
	2.0223	-0.0096
	2.7038	0.5953
	3.3853	1.5155
	3.7261	2.559
Fixed Inlet Pressure (kPa)	3.7261	2.559
	2.7791	0.8712
	1.8425	-0.0464
	1.3781	-0.9439
	1.1007	-1.0717

Looking at Table 2, it's interesting to see that force increases as pressure ratio increases. Pressure differences that are large will cause great acceleration of the flow through the nozzle and subsequently result in greater changes in momentum which causes a corresponding increase in the force exerted on the nozzle.

Table 3. Bernoulli's Equation

	Ratio of Inlet Pressure over Outlet Pressure	Percent Difference b/n LHS & RHS
Fixed Outlet Pressure (kPa)	3.7261	48.3889
	2.7791	49.6697
	1.8425	28.6328
	1.3781	1.7656
	1.1007	4.4095
Fixed Inlet Pressure (kPa)	1.3408	3.334
	2.0223	29.3543
	2.7038	50.2326
	3.3853	57.42
	3.7261	48.3889

Looking at Table 3, Bernoulli's equation, which relates the pressure, velocity, and height of fluid flow, does not work here. At higher pressure ratios, the difference between the LHS and RHS approaches almost 50%, clearly indicating that the discrepancy in values renders the use of Bernoulli's equation inappropriate here.

Table 4. Shock Formation in Nozzle

	Ratio of Inlet Pressure over Outlet Pressure	Theoretical Shock Location (In terms of Area)	Actual Shock based on Experimental Data
Fixed Outlet Pressure (kPa)	3.7261	1.94	1.855
	2.7791	1.42	1.68
	1.8425	No shock	No shock
	1.3781	No shock	No shock
	1.1007	No shock	No shock
Fixed Inlet Pressure (kPa)	1.3408	No shock	No shock
	2.0223	1.13	1.35
	2.7038	1.42	1.68
	3.3853	1.77	1.855
	3.7261	1.94	1.855

Looking at Table 4, shockwaves occur when the pressure ratio reaches about 2.0. The positions of the shockwaves compared to the area ratio for both experimental and theoretical configurations agree for all pressures that were tested; thus, it is concluded that shockwaves will not form at lower pressure ratios either experimentally or theoretically.

B. Experimental Plots

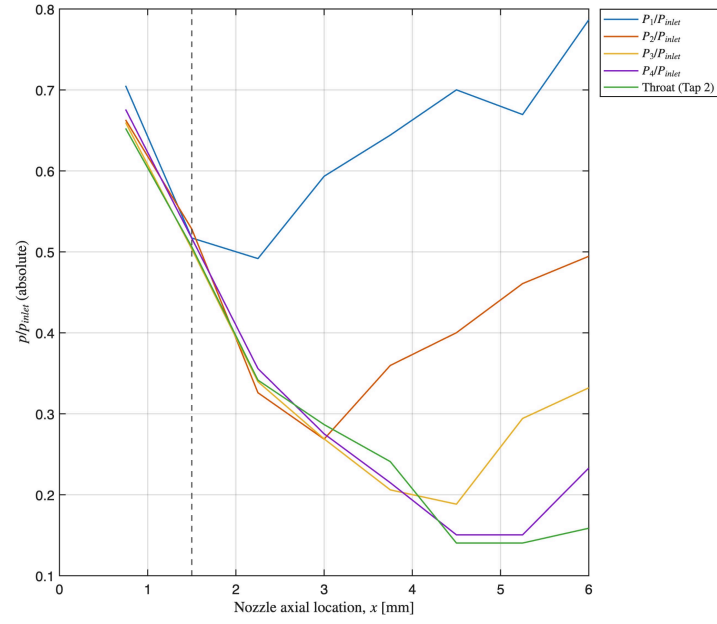


Fig. 4 The Effect of Nozzle Location on Pressure Ratios of a Fixed Inlet.

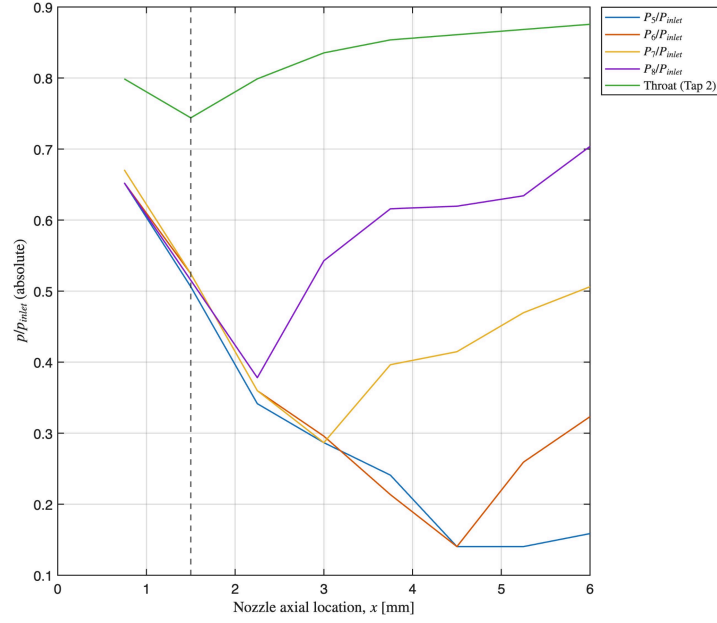


Fig. 5 The Effect of Nozzle Location on Pressure Ratios of a Fixed Outlet.

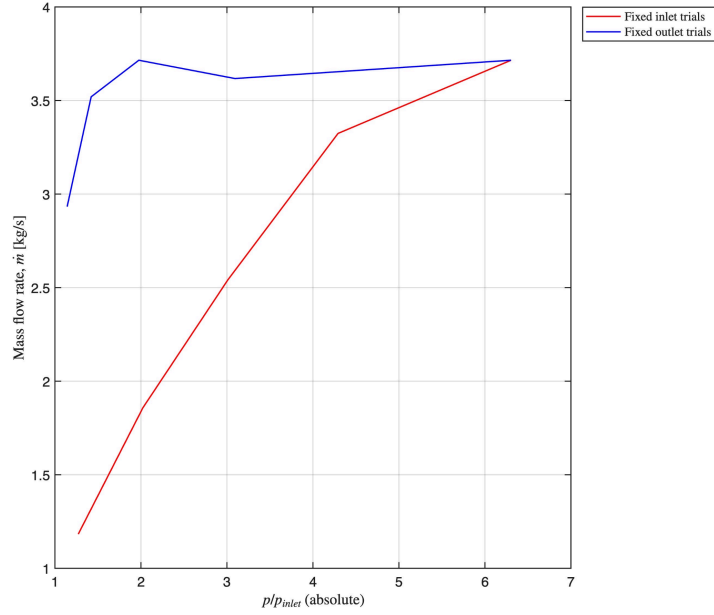


Fig. 6 The Effect of Absolute Pressure Ratios on Mass Flow.

These plots are for the experiment. From the first two plots, it is clear that all of the pressure conditions exhibit shockwaves, some of them clearer than others. It is also clear to see a sort of exponential decay effect of where the shocks occur over the various pressure ratios. Looking at Fig. 6, the nozzle is choked, and the critical pressure ratio in this experiment looks to be close to 2. It is clear that the nozzle chokes because both inlet and outlet values plateau and converge.

C. Theoretical Plots

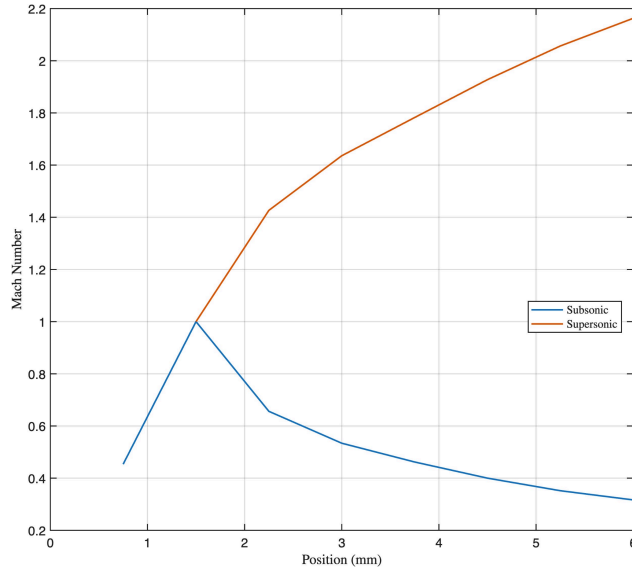


Fig. 7 The Effect of Position on Theoretical Mach Number.

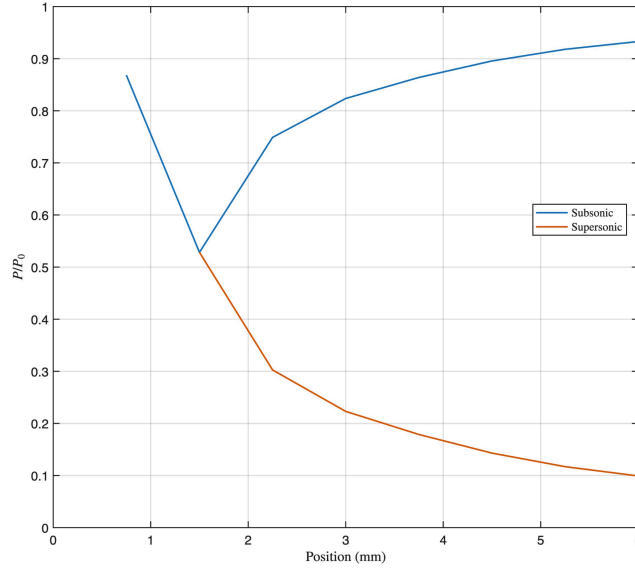


Fig. 8 The Effect of Position on Theoretical Pressure Ratio.

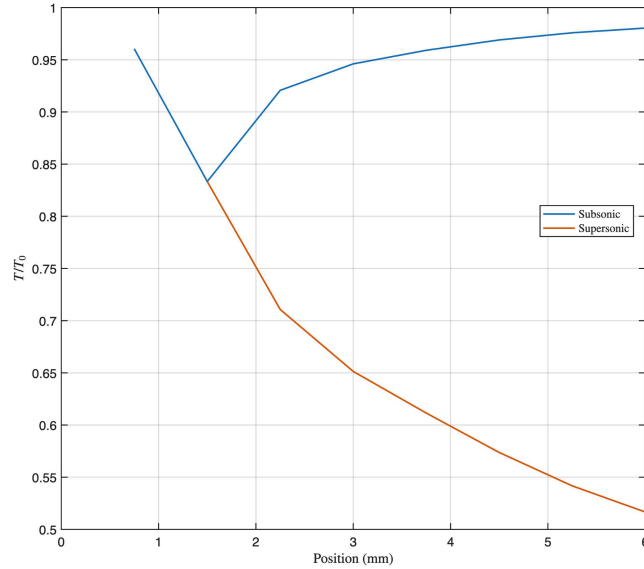


Fig. 9: The Effect of Position on the Theoretical Temperature Ratio.

There are two cases for each plot. The plots depict both subsonic and supersonic pathing scenarios. Looking at Fig. 7, the subsonic and supersonic plots both follow each other until the Mach number reaches a critical 1, and from there the supersonic Mach number continues to increase, while the subsonic peaks and proceeds to drop. As for Figures 8 and 9, they both follow the same trend. The pressure ratios and temperature ratios for the subsonic path restore back to the original values, while the supersonic values continue to grow (Mach number) or decay (pressure ratios and temperature ratios).

V. Conclusion

For the pressure ratio plots p_t/p_{in} vs nozzle position, the pressure should decrease when the flow accelerates through the converging section and reach a minimum near the throat as the Mach number becomes one. Past the throat, the pressure is dependent on whether the flow is subsonic or supersonic. When flow is subsonic, the pressure should begin to increase, but if flow is supersonic, the pressure will decrease as flow accelerates. The experimental p_t/p_{in} plots match this behavior. The pressure ratio decreases toward the throat as expected. Past the throat, for the subsonic case, pressure increases gradually throughout the diverging section. When airflow is supersonic, the pressure continues to decrease past the throat until it reaches a shock that adjusts the flow conditions to the outlet conditions. In summary, the pressure plots from the experimental results are consistent with the expected theory of flow through a converging-diverging nozzle.

Mass flow rate also acted as expected. In theory, the mass flow rate should increase with the pressure ratio until the flow hits the choking condition in the throat. When choked, the Mach number at the throat should be 1, $M = 1$ and the mass flow should hit its peak value. In the plot from the experiment, this trend is followed for the fixed inlet trials; mass flow rate increases as the pressure ratio increases until it approaches a max value of 3.7 kg/s . For the fixed outlet trials, the mass flow rate quickly moved to a constant value, proving the flow is choked.

In the experiment, the flow was compressible and could include shocks. However, when we evaluate Bernoulli's equation using the experimental data, we assume a steady, incompressible, and inviscid flow with no energy losses. With this assumption in mind, Bernoulli's is only expected to hold approximately in regions where the flow stays subsonic with small compressible effects. In the converging portion of the nozzle, Bernoulli's is reasonable due to being subsonic, but Bernoulli's is invalid when flow reaches supersonic conditions. As a result, the difference between the two sides of Bernoulli's equation is expected in the region.

The explanation for the forces on the nozzle is as the air moves through the nozzle, it will begin to accelerate, especially in the throat and the diverging section when the flow is supersonic. With the change in velocity comes a change in momentum of the fluid. This change in momentum produces a reaction force on the nozzle, as described by the conservation of momentum equation. The strongest forces occur from a large pressure difference between the inlet and outlet, since the pressure difference results in the biggest acceleration of air.

During the experiment, there are several possible sources of error. Measurement uncertainty in the pressure gauges and rotameter could definitely affect the accuracy of the pressure and mass flow rates recorded in the lab. Any small leaks in connection points to the pressure taps or any misalignment can result in uncertainty. The uncertainty also comes from the theoretical analysis because we assume the flow is perfect and simple, but in reality, it has friction and other effects that complicate it. All these factors may contribute to a difference from the theoretical values to the experimental data. To improve the experiment in future trials, precise pressure sensors can be added for more accurate data collection, and additional measures such as controlling the temperature and ensuring a more uniform flow profile can also enhance the reliability of the results.

Ultimately, the experimental results were meaningful and aligned closely with the expected behavior of compressible nozzle flow. The pressure distributions showed that the flow sped up as it got closer to the throat. This was because the flow was subsonic and supersonic in the diverging part of the tube. The experimental results for the mass flow rate confirmed choking occurred in the converging-diverging nozzle since the flow rate reached a maximum in the fixed outlet trials. The results of the experiment supported the theory of compressible flow, showing that converging and diverging nozzles can increase flow from subsonic to supersonic.

Appendix A

Table 1. Experimental Speed of Sound of Various Diaphragms.

Material	Speed of Sound (m/s)
Aluminum	~90.230
2-mil Mylar	~74.331
5-mil Mylar	~59.841

Table 2. Speed of Sound of Various Gases.

	Speed of Sound (m/s)
H ₂	~1,270
He ₂	~970
N ₂	~333.6
Air	~331

Table 3. Experimental Data

Material	P4 P1 Ratio	P2 P1 Ratio	Shock Speed [m/s]	Flow Speed [m/s]
Aluminum Foil	99.586	20.925	383.65	302.02
2-mil Mylar	162.9	43.59	455.22	369.24
5-mil Mylar	363.06	87.639	519.15	426.87

Table 4. Analytical Data

Material	P4 P1 Ratio	P2 P1 Ratio	Shock Speed [m/s]	Flow Speed [m/s]
Aluminum Foil	31.376	20.925	1466.4	1154.4
2-mil Mylar	66.196	43.59	2112.2	1713.2
5-mil Mylar	133.88	87.639	2992.1	2460.3

Appendix B

Density

$$\begin{aligned}
 PV &= nRT \\
 \rho &= P/RT \\
 P_{\text{atm}} &= 97.71 \text{ kPa} \\
 R &= 287 \text{ J/mol} \cdot \text{K} \\
 T_{\text{atm}} &= 22.9^\circ\text{C} = 22.9 + 273.15 = 296 \text{ K} \\
 \rho &= 97.71 / (287 * 296) \\
 \rho &= 0.00115 \text{ kg/m}^3
 \end{aligned}
 \tag{1}$$

Density Uncertainty

$$\delta\rho = \sqrt{\left(\frac{\delta P}{RT}\right)^2 + \left(\left(\frac{P\delta T}{RT^2}\right)\Delta T\right)^2}$$

$$P_{\text{atm}} = 97.71 \text{ kPa}$$

$$R = 287 \text{ J/mol}\cdot\text{K}$$

$$T_{\text{atm}} = 22.9 \text{ }^\circ\text{C} = 22.9 + 273.15 = 296\text{K}$$

$$\delta P = 0.1 \text{ kPa}$$

$$\delta T = 1 \text{ K}$$

$$\delta\rho = \sqrt{(0.1/287 * 1)^2 + ((97.71 * 1/287 * 1^2)1)^2}$$

$$\delta\rho = \pm 0.00029382 \text{ kg/m}^3$$

Appendix C

```

%% High Speed Aero Lab #1
clear; clc; close all
%% Import Givens
cel_data = {};
cel_data{1} = importdata("Wed_730_B1_A1.txt").data;
cel_data{2} = importdata("Wed_730_B1_2mil.txt").data;
cel_data{3} = importdata("Fri_200_A1_5mil.txt").data;
PSI_to_Pa = 6894.76; % 1 psi = 6894.76 Pa
mV_per_kPa = 3.378;
delx_PT1_2 = 48 + 1/32; % inches
delx_PT2_end = 36.25 + 1/32; % inches
PT_uncert = 1.3; % percent
Time_uncert = 2.1e-4; % seconds
Temp_uncert = 1; % Kelvin
Abs_P_uncert = 0.1; % kPa
P_atm = 97.71; % kPa
T_ambient = 22.9 + 273.15; % K
R = 287;
gamma = 1.4;
rho = P_atm*1000/(R*T_ambient);
rhoerr = sqrt( (1/(R*T_ambient)*Abs_P_uncert)^2 +
((-P_atm/(R*(T_ambient^2)))*Temp_uncert)^2 );
%% Data Organization
main_text_labels = ["A1", "mil2", "mil5"];
text_labels = ["driven_statP", "driver_statP", "DynP_1_V", "DynP_2_V"];
raw_data = struct;
for i = 1:length(cel_data)
    for j = 1:length(text_labels)
        raw_data.(main_text_labels(i)).(text_labels(j)) = cel_data{i}(:,j);
    end
end
%% Refine Givens
data = struct;
dt = 2.5e-5;
for i = 1:length(cel_data)

```

```

    for j = 1:length(text_labels)
        if j == 1
            data.(main_text_labels(i)).(text_labels(j)) =
(raw_data.(main_text_labels(i)).(text_labels(j)) * PSI_to_Pa)/1000; % kPa
        end
        if j == 2
            data.(main_text_labels(i)).(text_labels(j)) =
(raw_data.(main_text_labels(i)).(text_labels(j)) * PSI_to_Pa)/1000 + P_atm; %
kPa abs
        end
        if j == 3
            data.(main_text_labels(i)).(text_labels(j)) =
raw_data.(main_text_labels(i)).(text_labels(j)) * 1000 / mV_per_kPa; % kPa
        end
        if j == 4
            data.(main_text_labels(i)).(text_labels(j)) =
raw_data.(main_text_labels(i)).(text_labels(j)) * 1000 / mV_per_kPa; % kPa
        end
    end
    N = size(cel_data{i},1);
    data.(main_text_labels(i)).Time = (0:N-1)' * dt;
end
%% Initial Plotting
plot_titles = ["Static Pressure over time, Driven & Driver Sections
[Aluminium]", ...
               "Dynamic Pressure over time, Positions 1 & 2 [Aluminium]", ...
               "Static Pressure over time, Driven & Driver Sections [2-mil
Mylar]", ...
               "Dynamic Pressure over time, Positions 1 & 2 [2-mil Mylar]", ...
               "Static Pressure over time, Driven & Driver Sections [5-mil
Mylar]", ...
               "Dynamic Pressure over time, Positions 1 & 2 [5-mil Mylar]"];
Data_xlims = [1.755, 1.77; ...
              1.705, 1.725; ...
              1.21, 1.23];
fig_count = 0;
labFS = 16; % xlabel/ylabel size
axFS = 13; % tick size
for i = 1:length(cel_data)
    fig_count = fig_count + 1;
    figure(fig_count); hold on; grid on; box on
    a = plot(data.(main_text_labels(i)).Time,
data.(main_text_labels(i)).driven_statP, "r-", 'LineWidth', 1.2);
    a.DisplayName = "Driven Static Pressure";
    b = plot(data.(main_text_labels(i)).Time,
data.(main_text_labels(i)).driver_statP, "b-", 'LineWidth', 1.2);
    b.DisplayName = "Driver Static Pressure";
    title(plot_titles(2*i - 1))
    xlabel('$t\ \mathrm{[s]}$', 'Interpreter', 'latex', 'FontSize', labFS)

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ylabel('$P_{\mathrm{[kPa]}}$', 'Interpreter', 'latex', 'FontSize', labFS)
set(gca, 'FontSize', axFS)
legend('Location', 'best')
fig_count = fig_count + 1;
figure(fig_count); hold on; grid on; box on
a = plot(data.(main_text_labels(i)).Time,
data.(main_text_labels(i)).DynP_1_V, "r-", 'LineWidth', 1.2);
a.DisplayName = "Dynamic PT 1";
b = plot(data.(main_text_labels(i)).Time,
data.(main_text_labels(i)).DynP_2_V, "b-", 'LineWidth', 1.2);
b.DisplayName = "Dynamic PT 2";
title(plot_titles(2*i))
xlabel('$t\ \mathrm{[s]}$', 'Interpreter', 'latex', 'FontSize', labFS)
ylabel('$P_{\mathrm{[kPa]}}$', 'Interpreter', 'latex', 'FontSize', labFS)
xlim(Data_xlims(i,:))
set(gca, 'FontSize', axFS)
legend('Location', 'best')
end
% Delta Struct: [Al, 2mil, 5mil]
P = [1.61338, 33.75961; ...
      2.38011, 103.75; ...
      2.04263, 179.014];
t = [1.76370, 1.76052; ...
      1.71085, 1.70817; ...
      1.22022, 1.21787];
delta_P = [abs(P(1,1) - P(1,2)), ...
           abs(P(2,1) - P(2,2)), ...
           abs(P(3,1) - P(3,2))];
t_DPT = [abs(t(1,1) - t(1,2)), ...
         abs(t(2,1) - t(2,2)), ...
         abs(t(3,1) - t(3,2))];
%% Calculations
dx_m = delx_PT1_2 * 0.0254;
W_exp = [];
W_unc = [];
W_error = [];
a1_exp = [];
P2P1_exp = [];
up_exp = [];
P4P1_exp = [];
M_1 = [];
P2P1_ana = [];
W_ana = [];
P4P1_ana = [];
up_ana = [];
dt_s = 2.5e-5;
for i = 1:length(cel_data)
    lab = main_text_labels(i);
    Time = data.(lab).Time(:);

```

```

t1 = min(t(i,:));
t2 = max(t(i,:));
Dt = t2 - t1;
W_exp(i) = dx_m / Dt;
W_error(i) = sqrt( ((1./Dt).*Time_uncert).^2 + ((dx_m./(Dt.^2)).*0).^2 );
P1 = min(P(i,:));
P2 = max(P(i,:));
P2P1_exp(i) = P2 / P1;
a_exp(i) = W_exp(i) ./ sqrt( ((gamma+1)./(2*gamma)) .* (P2P1_exp(i) - 1) + 1
);
up_exp(i) = (a_exp(i)/gamma) * (P2P1_exp(i) - 1) * ...
            sqrt( (2*gamma/(gamma+1)) / (P2P1_exp(i) + (gamma-1)/(gamma+1))
);
[~, idx_t2] = min(abs(Time - t2));
P4 = max(data.(lab).driver_statP(1:idx_t2));
P4P1_exp(i) = P4 / P1;
a_ana = sqrt(gamma * R * T_ambient);
M_1(i) = sqrt( 1 + ((gamma+1)/(2*gamma)) * (P2P1_exp(i) - 1) );

P2P1_ana(i) = 1 + ((2*gamma)/(gamma+1)) * (M_1(i)^2 - 1);

W_ana(i) = a_ana * M_1(i);

P4P1_ana(i) = P2P1_ana(i) .* ...
            (1 - ((gamma-1).*(P2P1_ana(i)-1)) ./ ...
            ((2*gamma).*(2*gamma + (gamma+1).*(P2P1_ana(i)-1))) ) .^
((-2*gamma)/(gamma-1));

up_ana(i) = (a_ana/gamma) * (P2P1_ana(i) - 1) * ...
            sqrt( (2*gamma/(gamma+1)) / (P2P1_ana(i) + (gamma-1)/(gamma+1))
);
end
%% Plotting analytical vs experimental
color = ["black", "blue", "magenta"];
text_lab = [
    "Experimental Solution[A1]"      "Analytical Solution [A1]";
    "Experimental Solution[2-mil]"    "Analytical Solution [2-mil]";
    "Experimental Solution[5-mil]"    "Analytical Solution [5-mil]"
];
];
ms = 55;
labFS = 18;
axFS = 14;
legFS = 12;
for i = 1:length(cel_data)
    figure(7); hold on; grid on; box on
    scatter(P4P1_exp(i), W_exp(i), ms, color(i), "+", 'LineWidth', 2.0,
'DisplayName', text_lab(i,1))
    scatter(P4P1_ana(i), W_ana(i), ms, color(i), "o", "filled", 'LineWidth',
1.2, "DisplayName", text_lab(i,2))

```

```

xlabel('$\frac{P_4}{P_1}$','Interpreter','latex','FontSize',labFS)
ylabel('$W\ \mathrm{[m/s]}$','Interpreter','latex','FontSize',labFS)
title('Shock Speed vs Driver Pressure Ratio')
set(gca,'FontSize',axFS)
legend('Location','best','FontSize',legFS)
figure(8); hold on; grid on; box on
scatter(P4P1_exp(i), P2P1_exp(i), ms, color(i), "+", 'LineWidth', 2.0,
"DisplayName", text_lab(i,1))
scatter(P4P1_ana(i), P2P1_ana(i), ms, color(i), "o", "filled", 'LineWidth',
1.2, "DisplayName", text_lab(i,2))
xlabel('$\frac{P_4}{P_1}$','Interpreter','latex','FontSize',labFS)
ylabel('$\frac{P_2}{P_1}$','Interpreter','latex','FontSize',labFS)
title('Pressure Ratio vs Driver Ratio')
set(gca,'FontSize',axFS)
legend('Location','best','FontSize',legFS)
figure(9); hold on; grid on; box on
scatter(P2P1_exp(i), W_exp(i), ms, color(i), "+", 'LineWidth', 2.0,
"DisplayName", text_lab(i,1))
scatter(P2P1_ana(i), W_ana(i), ms, color(i), "o", "filled", 'LineWidth',
1.2, "DisplayName", text_lab(i,2))
xlabel('$\frac{P_2}{P_1}$','Interpreter','latex','FontSize',labFS)
ylabel('$W\ \mathrm{[m/s]}$','Interpreter','latex','FontSize',labFS)
title('Shock Speed vs Pressure Ratio')
set(gca,'FontSize',axFS)
legend('Location','best','FontSize',legFS)
figure(10); hold on; grid on; box on
scatter(P2P1_exp(i), up_exp(i), ms, color(i), "+", 'LineWidth', 2.0,
"DisplayName", text_lab(i,1))
scatter(P2P1_ana(i), up_ana(i), ms, color(i), "o", "filled", 'LineWidth',
1.2, "DisplayName", text_lab(i,2))
xlabel('$\frac{P_2}{P_1}$','Interpreter','latex','FontSize',labFS)
ylabel('$u_p\ \mathrm{[m/s]}$','Interpreter','latex','FontSize',labFS)
title('Particle Velocity vs Pressure Ratio')
set(gca,'FontSize',axFS)
legend('Location','best','FontSize',legFS)
end

```

Appendix D

Member Contributions:

Carolina Meza - Data Collection, Procedures, Appendix(es)

Bhuvan Vemuri - Results

Joseph Normandeau - Conclusion, Appendix(es)

Isaac Kaps - Code, Appendix(es)

Ahmed N Soliman - Introduction

Walter Moffatt - Editing/Corrections

Amber Wilderspin - Abstract, Nomenclature

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